FATE-H Statements

Formalization Contribution

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Mathematical Contribution

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Exercise (1). *Prove that if H is a subgroup of G of index n, then there is a normal subgroup K of G such that $K \leq H$ and $[G : K] \leq n!$*

```
import Mathlib

/--
Prove that if $H$ is a subgroup of $G$ of index $n$, then there is a normal subgroup $K$ of $G$
such that $K\leq H$ and $[G:K]\leq n!$
-/
theorem subgroup_normal_index_le_factorial {G : Type} [Group G] {n : ℕ} (hn : n ≠ 0)
    (H : Subgroup G) (hH : H.index = n) :
    ∃ K : Subgroup G, K.Normal ∧ K ≤ H ∧ K.index ≠ 0 ∧ K.index ≤ n.factorial := by
  sorry
```

Exercise (2). *Prove that if $\#G = 56$ then G is not simple.*

```
import Mathlib

/--
Prove that if $\#G = 56$ then $G$ is not simple.
-/
theorem not_isSimpleGroup_of_card_eq_56 {G : Type} [Group G] (hG : Nat.card G = 56) :
    ¬ IsSimpleGroup G := by
  sorry
```

Exercise (3). *Prove that if $\#G = 3393$ then G is not simple.*

```
import Mathlib

/--
Prove that if $\#G = 3393$ then $G$ is not simple.
-/
theorem not_isSimpleGroup_of_card_eq_3393 {G : Type} [Group G] (h : Nat.card G = 3393) : ¬
    IsSimpleGroup G := by
  sorry
```

Exercise (4). *Prove that if p is a prime and P is a non-abelian group of order p^3, then $|Z(P)| = p$ and $P/Z(P) \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.*

```
import Mathlib

/--
Prove that if $p$ is a prime and $P$ is a non-abelian group of order $p^3$, then $|Z(P)| = p$
and $P/Z(P) \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.
-/
theorem nonempty_mulEquiv_zMod_prod_zMod {p : ℕ} [Fact p.Prime] {P : Type} [Group P] (hp : Nat.card P
    = p ^ 3)
    (h : ∃ (a b : P), a * b ≠ b * a) : Nat.card (Subgroup.center P) = p ∧
    Nonempty ((P ⧸ Subgroup.center P) ≃* Multiplicative ((ZMod p) × (ZMod p))) := by
  sorry
```

Exercise (5). *Let R be a ring with $1 \neq 0$. For two elements $a, b \in R$, if $1 - ab$ is a unit, then $1 - ba$ is a unit.*

```
import Mathlib

/--
Let $R$ be a ring with $1 \neq 0$. For two elements $a,b \in R$, if $1-ab$ is a unit,
then $1-ba$ is a unit.
-/
theorem isUnit_one_sub_mul {R : Type} [Ring R] [Nontrivial R] {a b : R} (h : IsUnit (1 - a * b)) :
    IsUnit (1 - b * a) := by
  sorry
```

Exercise (6). *Show that a prime p can be written as $p = a^2 + ab + b^2$ with $a, b \in \mathbb{Z}$ if and only if $p = 3$ or $p \equiv 1 \pmod 3$.*

```
import Mathlib

/--
Show that a prime $p$ can be written as $p = a^2+ab+b^2$ with $a,b \in \mathbb{Z}$ if and only
if $p=3$ or $p \equiv 1 \pmod 3$.
-/
theorem exists_sum_two_squares_iff_mod_three_eq_one (p : ℕ) (hp : p.Prime) :
```

```
    (∃ a b : ℤ, a ^ 2 + a * b + b ^ 2 = p) ↔ p = 3 ∨ p % 3 = 1 := by
  sorry
```

Exercise (7). *Let G be a group and let $K \subseteq H$ be subgroups of G with $K \lhd H$. If $H \lhd G$ and $C_H(K) = 1$, prove that H centralizes $C_G(K)$.*

```
import Mathlib

open Subgroup

/--
Let $G$ be a group and let $K\subseteq H$ be subgroups of $G$ with $K \lhd H$.
   If $H \lhd G$ and $C_H(K)=1$, prove that $H$ centralizes $C_G(K)$.
-/
theorem le_centralizer_centralizer_of_centralizer_eq_bot {G : Type} [Group G] (H K : Subgroup G)
    [H.Normal] (h1 : K ≤ H) [(K.subgroupOf H).Normal]
    (h2 : Subgroup.centralizer (K.subgroupOf H) = (⊥ : Subgroup H)) :
    H ≤ Subgroup.centralizer (Subgroup.centralizer (K : Set G) : Set G) := by
  sorry
```

Exercise (8). *Show that if a Sylow 2-subgroup of G is nontrivial and cyclic, then G has a subgroup H with $[G : H] = 2$.*

```
import Mathlib

/--
Show that if a Sylow $2$-subgroup of $G$ is nontrivial and cyclic, then $G$ has a subgroup $H$
with $[G:H] =2$.
-/
theorem exists_index_two_of_sylow_two_isCyclic {G : Type} [Group G] [Finite G] (P : Sylow 2 G)
    (h1 : P.toSubgroup ≠ ⊥) [IsCyclic P] : ∃ H : Subgroup G, H.index = 2 := by
  sorry
```

Exercise (9). *If n is odd and $n \geq 3$, show that the identity is the only element of D_{2n} which commutes with all elements of D_{2n}.*

```
import Mathlib

/--
If \( n \) is odd and \( n \geq 3 \), show that the identity is the only element of
\( D_{2n} \) which commutes with all elements of \( D_{2n} \).
-/
theorem DihedralGroup.centralizer_eq_bot {n : ℕ} (hn : Odd n) (h : n ≥ 3) :
    Subgroup.centralizer τ = (⊥ : Subgroup (DihedralGroup n)) := by
  sorry
```

Exercise (10). *Determine the last two digits of $3^{3^{100}}$.*

```
import Mathlib

/--
Prove that the last two digits of $3^{3^{100}} is 03.
-/
theorem three_pow_three_pow_mod_100 : 3 ^ (3 ^ 100) % 100 = 3 := by
  sorry
```

Exercise (11). *Let G be a group of order 3825. Prove that if H is a normal subgroup of order 17 in G, then $H \leq Z(G)$.*

```
import Mathlib

/--
Let $G$ be a group of order $3825$. Prove that if $H$ is a normal subgroup of order $17$ in $G$,
then $H \leq Z(G)$.
-/
theorem le_center_of_card_eq_17_of_card_eq_3825 {G : Type} [Group G] (h : Nat.card G = 3825)
    (H : Subgroup G) [H.Normal] (hH : Nat.card H = 17) : H ≤ Subgroup.center G := by
  sorry
```

Exercise (12). *Prove that $SL_2(\mathbb{F}_3)/Z(SL_2(\mathbb{F}_3)) \leq A_4$.*

```
import Mathlib

open MatrixGroups

/--
Prove that \( SL_2(\mathbb{F}_3) / Z(SL_2(\mathbb{F}_3)) \le A_4 \).
-/
theorem exists_sl_quot_center_monoidHom_alternatingGroup :
    ∃ φ : SL(2,ZMod 3) / Subgroup.center SL(2,ZMod 3) →* alternatingGroup (Fin 4),
    Function.Injective φ := by
  sorry
```

Exercise (13). *Prove that the number of Sylow p-subgroups of $\mathrm{GL}_2(\mathbb{F}_p)$ is $p+1$.*

```
import Mathlib

open Matrix

/--
Prove that the number of Sylow $p$-subgroups of $\operatorname{GL}_2(\mathbb{F}_p)$ is $p + 1$.
-/
theorem card_sylow_gl_two_eq_add_one (p : ℕ) [Fact p.Prime] :
    Nat.card (Sylow p <| GL (Fin 2) (ZMod p)) = p + 1 := by
  sorry
```

Exercise (14). *Let S be any ring and let $n > 2$ be an integer. Prove that if A is any strictly upper triangular matrix in $M_n(S)$, then $A^n = 0$. (A strictly upper triangular matrix is one whose entries on and below the main diagonal are all zero.)*

```
import Mathlib

/--
Let $S$ be any ring and let $n>2$ be an integer.
Propose a proof that if $A$ is any strictly upper triangular matrix in $M_n(S)$, then $A^n = 0$.
(A strictly upper triangular matrix is one whose entries on and below the main diagonal are all
zero.)
-/
theorem pow_eq_zero_of_strictly_upper_triangle {S : Type} [Ring S] (n : ℕ) (hn : 2 < n)
    (A : Matrix (Fin n) (Fin n) S) (hA : ∀ (i : Fin n), ∀ (j : Fin n), i ≥ j → A i j = 0) :
    A ^ n = 0 := by
  sorry
```

Exercise (15). *Prove that the ring $\mathbb{Z}[\sqrt{-5}]$ is not a principal ideal domain.*

```
import Mathlib

/--
Prove that the ring $\mathbb{Z}[\sqrt{-5}]$ is not a principal ideal domain.
-/
theorem not_isPrincipalIdealRing_Zsqrtd_neg_five : ¬IsPrincipalIdealRing (Zsqrtd (-5)) := by
  sorry
```

Exercise (16). *Let R be an integral domain and let i, j be relatively prime integers. Prove that the ideal $(x^i - y^j)$ is a prime ideal in $R[x, y]$.*

```
import Mathlib

open MvPolynomial Ideal
/--
Let \( R \) be an integral domain and let \( i, j \) be relatively prime integers. Prove that the
    ideal \( (x^i - y^j) \) is a prime ideal in \( R[x, y] \).
-/
theorem span_pow_sub_pow_isPrime_of_coprime {R : Type} [CommRing R] [IsDomain R] {i j : ℕ}
    (hi : i > 0) (hj : j > 0) (h : Nat.Coprime i j) :
    (span {(X 0 ^ i - X 1 ^ j : MvPolynomial (Fin 2) R)}).IsPrime := by
  sorry
```

Exercise (17). *Prove that $\frac{x^p-1}{x-1}$ is irreducible in $\mathbb{Z}[x]$.*

```
import Mathlib

open Polynomial
```

```
/--
Prove that $\frac{x^{p}-1}{x-1}$ is irreducible in $\mathbb{Z}[x].$
-/
theorem irreducible_X_pow_p_sub_one_div_X_sub_one (p : ℕ) [hp : Fact (Nat.Prime p)] :
    Irreducible (((Polynomial.X : ℤ[X]) ^ p - 1) /ₘ (Polynomial.X - 1)) := by
  sorry
```

Exercise (18). *Prove that* $x^2 + y^2 - 1$ *is irreducible in* $\mathbb{Q}[x, y]$.

```
import Mathlib

open MvPolynomial

/--
Prove that \( x^2 + y^2 - 1 \) is irreducible in \( \mathbb{Q}[x, y] \).
-/
theorem irreducible_pow_two_add_pow_two_sub_one :
    Irreducible ((X 0) ^ 2 + (X 1) ^ 2 - 1 : MvPolynomial (Fin 2) ℚ) := by
  sorry
```

Exercise (19). *Prove that any finite group is isomorphic to a subgroup of* A_n *for some* n.

```
import Mathlib

/--
Prove that any finite group is isomorphic to a subgroup of $A_n$ for some $n$.
-/
theorem exists_subgroup_alternatingGroup_mulEquiv {G : Type} [Group G] [Finite G] :
    ∃ (n : ℕ) (H : Subgroup (alternatingGroup (Fin n))), Nonempty (G ≃* H) := by
  sorry
```

Exercise (20). *Prove that if* G *is a nonabelian group of order* p^3 *(*p *prime), then the center of* G *is the subgroup generated by all elements of the form* $aba^{-1}b^{-1}$ *(*$a, b \in G$*).*

```
import Mathlib

/--
Prove that if $G$ is a nonabelian group of order $p^3$ ($p$ prime),
    then the center of $G$ is the subgroup generated by all elements of
    the form $aba^{-1}b^{-1}$ ($a, b \in G$).
-/
theorem center_eq_commutator_of_pow_three_eq_card {G : Type} [Group G] {p : ℕ} [hp : Fact p.Prime]
    (hG : p ^ 3 = Nat.card G) (h : ∃ x y : G, x * y ≠ y * x) :
    Subgroup.center G = commutator G := by
  sorry
```

Exercise (21). *Prove that the order of* $\mathrm{Aut}(\mathbb{Z}_3 \times \mathbb{Z}_9)$ *is 108.*

```
import Mathlib

/--
Prove that the order of Aut(Z_3 × Z_9) is 108.
-/
theorem card_addAut_eq_108 : Nat.card (AddAut <| ZMod 3 × ZMod 9) = 108 := by
  sorry
```

Exercise (22). *Let D_8 be the dihedral group with 8 elements. Prove that* $\mathrm{Aut}(D_8) \cong D_8$.

```
import Mathlib

/--
Let $D_8$ be the dihedral group with $8$ elements. Prove that $\mathrm{Aut}(D_8) \cong D_8$.
-/
theorem nonempty_mulAut_dihedralGroup_four : Nonempty (MulAut (DihedralGroup 4) ≃* DihedralGroup 4)
     := by
  sorry
```

Exercise (23). *Prove that if $\#G = 1053$ then G is not simple.*

```
import Mathlib

/--
Prove that if $\#G = 1053$ then $G$ is not simple.
-/
theorem not_isSimpleGroup_of_card_eq_1053 (G : Type) [Group G]
    [Finite G] (h_card : Nat.card G = 1053) : ¬ IsSimpleGroup G := by
  sorry
```

Exercise (24). *Prove that $\mathbb{Q}/\mathbb{Z}$ has no proper subgroups of finite index.*

```
import Mathlib

/--
Prove that $\mathbb{Q}/\mathbb{Z}$ has no proper subgroups of finite index.
-/
theorem eq_top_of_finiteIndex (H : AddSubgroup (ℚ / (Int.castAddHom ℚ).range)) (h_fin :
     H.FiniteIndex) :
    H = ⊤ := by
  sorry
```

Exercise (25). *Let $R = \mathbb{Z} + x\mathbb{Q}[x] \subset \mathbb{Q}[x]$be the set of polynomials in x with rational coefficients whose constant term is an integer. Prove that R is not a U.F.D.*

```
import Mathlib

open Polynomial
/--
Let $R=\mathbb{Z}+x\mathbb{Q}[x]\subset \mathbb{Q}[x]$be the set of polynomials in x with rational
coefficients whose constant term is an integer. Prove that $R$ is not a U.F.D.
-/
theorem not_uniqueFactorizationMonoid_adjoin_int : ¬ UniqueFactorizationMonoid
    (Algebra.adjoin ℤ ({f | ∃ h : ℚ[X], f = X * h} : Set ℚ[X])) := by
  sorry
```

Exercise (26). *Suppose G is a group and H is a maximal subgroup of G. Show that either $Z(G) \leq H$ or $[G,G] \leq H$. (A maximal subgroup contains either the center or the commutator subgroup.)*

```
import Mathlib

/--
Suppose \( G \) is a group and \( H \) is a maximal subgroup of \( G \). Show that either \( Z(G)
\leq H \) or \( [G,G] \leq H \). (A maximal subgroup contains either the center or the commutator
 subgroup.)
-/
theorem center_le_or_commutator_le_of_isCoatom {G : Type} [Group G] (H : Subgroup G)
    (h : IsCoatom H) : Subgroup.center G ≤ H ∨ commutator G ≤ H := by
  sorry
```

Exercise (27). *Let F be a field contained in the ring of $n \times n$ matrices over $\mathbb{Q}$. Prove that $[F : \mathbb{Q}] \leq n$.*

```
import Mathlib

/--
Let $F$ be a field contained in the ring of $n \times n$ matrices over $\mathbb{Q}$.
Prove that $[F:\mathbb{Q}] \leq n$.
-/
theorem rank_le_of_subalgebra_matrix {n : ℕ} (F : Subalgebra ℚ (Matrix (Fin n) (Fin n) ℚ))
    (h : IsField F) : Module.rank ℚ F ≤ n := by
  sorry
```

Exercise (28). *Let k be a perfect field of characteristic $p > 0$. Let $F = k(t)$ be the field of rational functions in one variable over k. Show that every finite extension E of F can be generated by one element, that is, there exists $\alpha \in E$ such that $E = F(\alpha)$.*

```
import Mathlib

open IntermediateField
```

```
/--
Let $k$ be a perfect field of characteristic $p > 0$.
Let $F = k(t)$ be the field of rational functions in one variable over $k$.
Show that every finite extension $E$ of $F$ can be generated by one element, that is,
there exists $\alpha \in E$ such that $E = F(\alpha)$.
-/
theorem exists_ratFunc_adjoin_eq_top {k : Type} [Field k] [PerfectField k] {p : ℕ} [Fact p.Prime]
    [CharP k p]
    {E : Type} [Field E] [Algebra (RatFunc k) E] [FiniteDimensional (RatFunc k) E] :
    ∃ α : E, (RatFunc k)⟮α⟯ = ⊤ := by
  sorry
```

Exercise (29). *Show that if F has characteristic p, then all degree p Galois extension of F are obtained by adjoining a zero of $x^p - x - a$ for some $a \in F$.*

```
import Mathlib

open IntermediateField Polynomial

/--
Show that if $F$ has characteristic $p$, then all degree $p$ Galois extension of $F$ is to
adjoin a zero of $x^p-x-a$ for some $a \in F$.
-/
theorem exists_nonempty_adjoin_root_X_pow_p_sub_X_sub_C
    {F E : Type} [Field F] {p : ℕ} [Fact p.Prime] [CharP F p] [Field E]
    [Algebra F E] [IsGalois F E] (h_deg : Module.finrank F E = p) :
    ∃ a : F, Nonempty (AdjoinRoot (X ^ p - X - C a : F[X]) ≃+* E) := by
  sorry
```

Exercise (30). *Let E be the splitting field of*

$$f(x) = \frac{x^7 - 1}{x - 1} = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$$

over $\mathbb{Q}$. Let ζ be a zero of $f(x)$, i.e., a primitive seventh root of 1. Let $\beta = \zeta + \zeta^2 + \zeta^4$. Show that the intermediate field $\mathbb{Q}(\beta)$ is actually $\mathbb{Q}(\sqrt{-7})$.

```
import Mathlib

open Polynomial
open scoped IntermediateField

/--
Let $E$ be the splitting field of
\[
f(x) = \frac{x^7 - 1}{x - 1} = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1
\]
over $\mathbb{Q}$. Let $\zeta$ be a zero of $f(x)$, i.e., a primitive seventh root of $1$.
```

```
Let $\beta = \zeta + \zeta^2 + \zeta^4$. Show that the intermediate field $\mathbb{Q}(\beta)$
is actually $\mathbb{Q}(\sqrt{-7})$.
-/
theorem nonempty_ringEquiv_adjoin_pow_two_add_seven {E : Type} [Field E] [Algebra ℚ E]
    [IsCyclotomicExtension {7} ℚ E] (ζ : E)
    (h : IsPrimitiveRoot ζ 7) (β : E) (hb : β = ζ + ζ ^ 2 + ζ ^ 4) :
    Nonempty (ℚ[β] ≃+* AdjoinRoot (X ^ 2 + 7 : ℚ[X])) := by
  sorry
```

Exercise (31). *Prove that the primitive n^{th} roots of unity form a basis over $\mathbb{Q}$ for the cyclotomic field of nth roots of unity if and only if n is squarefree.*

```
import Mathlib

/--
Prove that the primitive $n^{\textrm{th}}$ roots of unity form a basis over $\mathbb{Q}$ for
the cyclotomic field of $n^{\textrm{th}}$ roots of unity if and only if $n$ is squarefree.
-/
theorem exists_basis_primitiveRoots_iff_squarefree {n : ℕ+} :
    (∃ b : Basis (primitiveRoots n (CyclotomicField n ℚ)) ℚ (CyclotomicField n ℚ),
    (b : _ → _) = (↑)) ↔ Squarefree n := by
  sorry
```

Exercise (32). *Prove that the splitting field of $x^4 - 2x^2 - 2$ over $\mathbb{Q}$ is of degree 8 with dihedral Galois group*

```
import Mathlib

open Polynomial

/--
The Galois group of the splitting field of $x^4 - 2x^2 - 2$ over $\mathbb{Q}$ is the
dihedral group with eight elements
-/
theorem nonempty_galois_mulEquiv_dihedralGroup_four :
    Nonempty (Gal (X ^ 4 - 2 * X ^ 2 - 2 : ℚ[X]) ≃* DihedralGroup 4) := by
  sorry
```

Exercise (33). *Let $f(x) \in \mathbb{Q}[x]$ be a polynomial of degree n $(n > 4)$ and the splitting field E of $f(x)$ has Galois group S_n over $\mathbb{Q}$. Let α be a zero of $f(x)$ in E. Prove that for any other root β of $f(x)$, there are precisely $(n-1)!$ elements in $\mathrm{Gal}(E/\mathbb{Q})$ that takes α to β.*

```
import Mathlib

open Polynomial

/--
```

```
Let $f(x) \in \mathbb{Q}[x]$ be a polynomial of degree $n$ ($n>4$) and the splitting field $E$
of $f(x)$ has Galois group $S_n$ over $\mathbb{Q}$. Let $\alpha$ be a zero of $f(x)$ in $E$.
Prove that for any other root $\beta$ of $f(x)$, there are precisely $(n-1)!$ elements in
$\mathrm{Gal}(E/\mathbb{Q})$ that takes $\alpha $ to $\beta$
-/
theorem card_gal_map_eq_eq_factorial {n : Nat} (hn : n > 4) (f : ℚ[X]) (hf : f.degree = n)
    (hf' : Irreducible f) (h : f.Gal ≃* (Equiv.Perm <| Fin n))
    {a b : SplittingField f} (ha : f.aeval a = 0) (hb : f.aeval b = 0) (neq : a ≠ b) :
    Nat.card {h : f.Gal // h a = b} = Nat.factorial (n - 1) := by
  sorry
```

Exercise (34). *Let E be a field of characteristic zero. Consider a prime q and an element $b \in E^\times$ that isn' t a q-th power. Let $E' = E(a)$ with $a^q = b$ and $E' \neq E$. Show that $X^q - b$ is reducible over E if and only if $[E' : E] < q$.*

```
import Mathlib

open Polynomial
/--
Let \( E \) be a field of characteristic zero. Consider a prime \( q \) and an element \( b
\in E^\times \) that isn' t a \( q \)-th power. Let \( E' = E(a) \) with \( a^q = b \). Show that
\( X^q - b \) is reducible over \( E \) if and only if \( [E' : E] < q \)
-/
theorem not_irreducible_iff_finrank_lt {E E': Type} [Field E] [CharZero E] [Field E'] [Algebra E E']
    {q : ℕ}
    [Fact q.Prime] {b : E} (hb : b ≠ 0) (not_pow : ∀ c : E, c ^ q ≠ b) {a : E'}
    (ha : a ^ q = algebraMap E E' b) (haE : ⊤ = IntermediateField.adjoin E {a}) :
    ¬ Irreducible (X ^ q - C b) ↔ Module.finrank E E' < q := by
  sorry
```

Exercise (35). *Let $D \in \mathbb{Z}$ be a squarefree integer and let $a \in \mathbb{Q}$ be a nonzero rational number. Show that $\mathbb{Q}(\sqrt{a\sqrt{D}})$ cannot be a cyclic extension of degree 4 over $\mathbb{Q}$.*

```
import Mathlib

open Polynomial IntermediateField

/--
Let $D \in \mathbb{Z}$ be a squarefree integer and let $a \in \mathbb{Q}$ be a nonzero rational
number. Show that $\mathbb{Q}(\sqrt{a\sqrt D})$ cannot be a cyclic extension of degree $4$ over
$\mathbb{Q}$.
-/
theorem isEmpty_adjoinRoot_zMod_four {d : ℤ} (hd : Squarefree d) {a : ℚ} (ha : a ≠ 0) :
    IsEmpty ((AdjoinRoot ((a⁻¹ • X ^ 2) ^ 2 - C (d : ℚ)) ≃ₐ[ℚ]
      AdjoinRoot ((a⁻¹ • X ^ 2) ^ 2 - C (d : ℚ))) ≃* Multiplicative (ZMod 4)) := by
  sorry
```

Exercise (36). *Let $f(X) = X^4 - X^2 - 1 \in \mathbb{Q}[X]$, K is the splitting field of f over $\mathbb{Q}$, prove that the number of intermediate fields of $K/\mathbb{Q}$ is 10.*

```
import Mathlib

open Polynomial IntermediateField

/--
Let $f(X)=X^4-X^2-1\in \mathbb{Q}[X]$, $K$ is the splitting field of $f$ over $\mathbb{Q}$,
prove that the number of intermediate fields of $K/\mathbb{Q}$ is 10.
-/
theorem card_intermediateField_splittingField_eq_ten :
    Nat.card (IntermediateField ℚ (X ^ 4 - X ^ 2 - 1 : ℚ[X]).SplittingField) = 10 := by
  sorry
```

Exercise (37). *Let L/K be a Galois extension of fields such that $\mathrm{Gal}(L/K)$ is cyclic of order n, generated by σ. Write $n = ab$ with $\gcd(a, b) = 1$. Let F_1 be the fixed field of σ^a and F_2 be the fixed field of σ^b. Suppose that $F_1 = K(\alpha)$ and $F_2 = K(\beta)$. Prove that $L = K(\alpha + \beta)$.*

```
import Mathlib

open Polynomial IntermediateField AdjoinRoot

/--
Let $L/K$ be a Galois extension of fields such that $\mathrm{Gal}(L/K)$ is cyclic of order $n$,
generated by $\sigma$. Write $n = ab$ with $\gcd(a,b) =1$. Let $F_1$ be the fixed field of
$\sigma^a$ and $F_2$ be the fixed field of $\sigma^b$. Suppose that $F_1 = K(\alpha)$ and
$F_2 = K(\beta)$.  Prove that $L = K(\alpha+\beta)$.
-/
theorem adjoin_add_eq_top_of_fixedField_zpowers {K L : Type} [Field K]
    [Field L] [Algebra K L] [IsGalois K L] (n a b : ℕ)
    (σ : L ≃ₐ[K] L) (hσ : orderOf σ = n) (cycle : Subgroup.zpowers σ = ⊤) (hn : n > 0)
    (hn' : n = a * b) (hab : Nat.Coprime a b) {α β : L}
    (hα : K⟮α⟯ = IntermediateField.fixedField (Subgroup.zpowers (σ ^ a)))
    (hβ : K⟮β⟯ = IntermediateField.fixedField (Subgroup.zpowers (σ ^ b))) : K⟮α + β⟯ = ⊤ := by
  sorry
```

Exercise (38). *Let p be a prime number and let F be a field containing p-th roots of unity. Let K be a Galois extension of F with Galois group $\mathbb{Z}/p \times \mathbb{Z}/p$. Show that there exist two elements $\alpha, \beta \in K^\times$ such that $K = F(\alpha, \beta)$ and $a = \alpha^p, b = \beta^p \in F$.*

```
import Mathlib

open IntermediateField

/--
Let $p$ be a prime number and let $F$ be a field containing $p$-th roots of unity.
Let $K$ be a Galois extension of $F$ with Galois group $\mathbb{Z}_p \times \mathbb{Z}_p$.
Show that there exist two elements $\alpha, \beta \in K^\times$ such that
$K= F(\alpha, \beta)$ and $a = \alpha^p, b = \beta^p \in F$.
-/
```

```
theorem exists_pow_p_mem_algebraMap_and_adjoin_eq_top {p : Nat} [Fact p.Prime]
    {F K : Type} [Field F] (hF : (primitiveRoots p F).Nonempty) [Field K] [Algebra F K]
    [IsGalois F K] (hK : (K ≃ₐ[F] K) ≃* Multiplicative (ZMod p × ZMod p)) :
    ∃ (α β : K), α ≠ 0 ∧ β ≠ 0 ∧ α ^ p ∈ (algebraMap F K).range ∧ β ^ p ∈ (algebraMap F K).range ∧
    IntermediateField.adjoin F {α, β} = ⊤ := by
  sorry
```

Exercise (39). *Prove that a splitting field of $X^{15} - 2$ over $\mathbb{F}_7$ is generated by a primitive 45-th root of unity.*

```
import Mathlib

open Polynomial
/--
Prove that a splitting field of \( X^{15} - 2 \) over \( \mathbb{F}_7 \) is generated by a
 primitive \( 45 \)-th root of unity.
-/
theorem exists_isPrimitiveRoot_and_adjoin_eq_top :
    letI : Fact (Nat.Prime 7) := by decide
    ∃ ζ : (X ^ 15 - 2 : (ZMod 7)[X]).SplittingField, IsPrimitiveRoot ζ 45 ∧
    IntermediateField.adjoin (ZMod 7) {ζ} = ⊤ := by
  sorry
```

Exercise (40). *Let K be the splitting field of an irreducible quintic polynomial $f(x) \in \mathbb{Q}[x]$ and let $\{\alpha_1, \ldots, \alpha_5\}$ be zeros of $f(x)$ in K. Show that if $\mathbb{Q}(\alpha_1, \alpha_2, \alpha_3) \neq K$, then $\mathrm{Gal}(K/\mathbb{Q}) \cong S_5$.*

```
import Mathlib

open Polynomial

/--
Let $K$ be the splitting field of a irreducible quintic polynomial $f(x) \in \mathbb{Q} [x]$
and let $\{\alpha_1, \dots, \alpha_5\}$ be zeros of $f(x)$ in $K$. Show that if $\mathbb{Q}
(\alpha_1, \alpha_2, \alpha_3) \neq K$, then $\\mathrm{Gal}(K/\mathbb{Q}) \cong S_5$.
-/
theorem nonempty_gal_mulEquiv_perm_fin_five {f : ℚ[X]} {K : Type} [Field K] [Algebra ℚ K]
    [IsSplittingField ℚ K f] (hf1 : Irreducible f) (hf2 : f.natDegree = 5) (a₁ a₂ a₃ : K)
    (ha1 : a₁ ∈ rootSet f K ∧ a₂ ∈ rootSet f K ∧ a₃ ∈ rootSet f K)
    (ha2 : a₁ ≠ a₂ ∧ a₂ ≠ a₃ ∧ a₃ ≠ a₁)
    (h : IntermediateField.adjoin ℚ {a₁, a₂, a₃} ≠ ⊤) :
    Nonempty (f.Gal ≃* (Equiv.Perm <| Fin 5)) := by
  sorry
```

Exercise (41). *Let p be a prime integer. Suppose that the degree of every finite extension of a field F is divisible by p. Prove that the degree of every finite extension of F is a power of p.*

```
import Mathlib

/--
Let $p$ be a prime integer. Suppose that the degree of every finite extension of a field $F$
 is divisible by $p$. Prove that the degree of every finite extension of $F$ is a power of $p$.
-/
theorem exists_finrank_eq_pow_of_dvd_finrank {F : Type} [Field F] (p : ℕ) [Fact (Nat.Prime p)]
    (h : ∀ (E : Type) [Field E] [Algebra F E] [FiniteDimensional F E],
    Module.finrank F E ≠ 1 → p ∣ Module.finrank F E)
    (E : Type) [Field E] [Algebra F E] [FiniteDimensional F E] :
    ∃ k, Module.finrank F E = p ^ k := by
  sorry
```

Exercise (42). *Let K be a field with $\mathrm{char}(K) \neq 2$. Consider Galois extensions L/K with $\mathrm{Gal}(L/K) \cong (\mathbb{Z}/2\mathbb{Z})^2$. Let $c \in L^\times$ be a nonsquare, and let $E = L(\sqrt{c})$. Prove that E is Galois over K if and only if for each $\sigma \in \mathrm{Gal}(L/K)$, the ratio $\sigma(c)/c$ is a square in L.*

```
import Mathlib

open IntermediateField AdjoinRoot Polynomial

/--
Let \( K \) be a field with \( \operatorname{char}(K) \neq 2 \). Consider Galois extensions
\( L/K \) with \( \operatorname{Gal}(L/K) \cong (\mathbb{Z}/2\mathbb{Z})^2 \). Let \( c \in L^\times
\) be a nonsquare, and let \( E = K(\sqrt{c}) \). Prove that \( E \) is Galois over \( K \) if and
only if for each \( \sigma \in \operatorname{Gal}(L/K) \), the ratio \( \sigma(c)/c \) is a square
in \( L \).
-/
theorem isGalois_adjoin_iff_isSquare (K L : Type) [Field K] [Field L] (h : ¬ CharP K 2) [Algebra K L]
    [IsGalois K L]
    (f : (L ≃ₐ[K] L) ≃* Multiplicative (ZMod 2 × ZMod 2)) (c : Lˣ) (hc : c.1 ≠ 0)
    (hcs : ¬ IsSquare c.1) [Fact (Irreducible (X ^ 2 - C c.1))] :
    IsGalois K (AdjoinRoot (X ^ 2 - C c.1)) ↔ ∀ σ : (L ≃ₐ[K] L), IsSquare (σ c / c) := by
  sorry
```

Exercise (43). *Let F be a field with $\mathbb{Q} \subseteq F \subseteq \mathbb{C}$, where $F/\mathbb{Q}$ is a finite* abelian *Galois extension. Let $\alpha \in F$ and let $f(x) \in \mathbb{Q}[x]$ be its minimal monic polynomial. Assume that $|\alpha| = 1$. Prove that $|\beta| = 1$ for every complex root β of $f(x)$.*

```
import Mathlib

open Polynomial

/--
Let $F$ be a field with $\mathbb{Q} \subseteq F \subseteq \mathbb{C}$, where $F/\mathbb{Q}$ is a
finite \emph{abelian} Galois extension. Let $\alpha \in F$ and let $f(x) \in \mathbb{Q}[x]$ be its
minimal monic polynomial. Assume that $|\alpha| =1$. Prove that $|\beta| = 1$ for every complex
root $\beta$ of $f(x)$.
```

```
-/
theorem norm_eq_one_of_mem_rootSet (F : IntermediateField ℚ ℂ) [FiniteDimensional ℚ F] [IsGalois ℚ F]
    (h : ∀ f g : F ≃ₐ[ℚ] F, f * g = g * f) (α : F) (f : ℚ[X])
    (h_min : f = minpoly ℚ α) (ha : ‖α‖ = 1)
    (β : ℂ) (hb : β ∈ f.rootSet ℂ) : ‖β‖ = 1 := by
  sorry
```

Exercise (44). *Let k be a finite field of size q. Show that the number of degree-19 monic irreducible polynomials over k is $\frac{q^{19}-q}{19}$.*

```
import Mathlib

open Polynomial

/--
Let $k$ be a finite field of size $q$. Show that the number of degree-$19$ monic irreducible
polynomials over $k$ is $\frac{q^{19} - q}{19}$.
-/
theorem card_monic_and_irreducible_and_natDegree_eq_19 {F : Type} (q : ℕ) [Field F]
    [Fintype F] (hF : Fintype.card F = q) :
    Nat.card { P : F[X] | P.Monic ∧ Irreducible P ∧ P.natDegree = 19 } =
    (q ^ 19 - q) / 19 := by
  sorry
```

Exercise (45). *Let F be a field, and let $f, g \in F[x] \setminus \{0\}$ be relatively prime and not both constant. Show that $F(x)$ has finite degree $d = \max(\deg(f), \deg(g))$ over its subfield $F\left(\frac{f}{g}\right)$.*

```
import Mathlib

open IntermediateField Polynomial

/--
Let $F$ be a field, and let $f, g \in F[x] \setminus \{0\}$ be relatively prime and not both
constant. Show that $F(x)$ has finite degree $d = \max(\deg(f), \deg(g))$ over its subfield $F\left
(\frac{f}{g}\right)$.
-/
theorem finrank_adjoin_dvd_eq_max_natDegree (F : Type) [Field F] (f g : F[X]) (hf : f ≠ 0)
    (hg : g ≠ 0) (hcoprime : IsCoprime f g) (hfg : ¬(f.natDegree = 0 ∧ g.natDegree = 0)) :
    Module.finrank F⟮(f : RatFunc F) / g⟯ (RatFunc F) = max f.natDegree g.natDegree := by
  sorry
```

Exercise (46). *Let α be a non-zero complex number such that $\alpha + \alpha^{-1}$ is contained in a quadratic number field. Let L be the normal closure of $\mathbb{Q}(\alpha)$. Show that $[L : \mathbb{Q}]$ divides 8.*

```
import Mathlib

open IntermediateField
```

```
/--
Let \( \alpha \) be a non-zero complex number such that \( \alpha + \alpha^{-1} \) is contained
in a quadratic number field. Let \( L \) be the normal closure of \( \mathbb{Q}(\alpha) \). Show
that \( [L : \mathbb{Q}] \) divides \( 8 \).
-/
theorem finrank_dvd_eight_of_add_inv_eq_algebraMap {α : ℂ} {K L : Type} [Field K] [NumberField K]
    [Algebra K ℂ] (hk : Module.finrank ℚ K = 2) {a : K} (h : α + α⁻¹ = algebraMap K ℂ a)
    [Field L] [Algebra ℚ L] [IsNormalClosure ℚ ℚ⟮a⟯ L] : Module.finrank ℚ L ∣ 8 := by
  sorry
```

Exercise (47). *Let ζ be a primitive 9-th root of unity in $\mathbb{C}$, so $\zeta^9 = 1$, and let $\omega = \zeta^3$ be a primitive 3-rd root of unity, so $\omega^3 = 1$. If $\alpha^3 = 3$, show that α is not a cube in $\mathbb{Q}(\zeta)$.*

```
import Mathlib

open Algebra

open scoped IntermediateField

/--
Let $\zeta$ be a primitive $9$-th root of unity in $\mathbb{C}$, so $\zeta^9 = 1$, and let
$\omega = \zeta^3$ be a primitive $3$-rd root of unity, so $\omega^3 = 1$. If $\alpha^3 = 3$,
show that $\alpha$ is not a cube in $\mathbb{Q}(\zeta)$.
-/
theorem not_exists_eq_pow_three_of_pow_three_eq_three (α : ℂ) (hα : α ^ 3 = 3) (ζ : ℂ)
    (hζ : IsPrimitiveRoot ζ 9) : ¬ ∃ β : ℚ⟮ζ⟯, α = β ^ 3 := by
  sorry
```

Exercise (48). *Prove that the cardinality $\mathrm{Aut}(\mathbb{C})$ (i.e. the group of field automorphism of $\mathbb{C}$) is infinite.*

```
import Mathlib

/--
Prove that the cardinality $\mathrm{Aut}(\CC)$ (i.e. the group of field automorphism of $\CC$)
is infinite.
-/
theorem infinite_complex_algEquiv : Infinite (ℂ ≃ₐ[ℚ] ℂ) := by
  sorry
```

Exercise (49). *Prove that every finitely generated extension of $\mathbb{Q}$ can be embeded into $\mathbb{C}$.*

```
import Mathlib

/--
Prove that every finitely generated extension of $\mathbb{Q}$ can be embeded into $\mathbb{C}$.
-/
theorem exists_algHom_complex_injective {F : Type} [Field F] [Algebra ℚ F]
    (h : (τ : IntermediateField ℚ F).FG) : ∃ f : F →ₐ[ℚ] ℂ, Function.Injective f := by
  sorry
```

Exercise (50). *Let m be a maximal ideal of $\mathbb{Z}[x_1, \ldots, x_n]$ and $F = \mathbb{Z}[x_1, \ldots, x_n]/m$. Show that F cannot have characteristic 0.*

```
import Mathlib

open MvPolynomial

/--
Let $m$ be a maximal ideal of $\mathbb{Z}[x_1, \dots, x_n]$ and $F = \mathbb{Z}[x_1, \dots, x_n]
/m$. Show that $F$ cannot have characteristic $0$.
-/
theorem not_charZero_mvPolynomial_quot {n : ℕ} (m : Ideal (MvPolynomial (Fin n) ℤ))
    [hm : m.IsMaximal] : ¬ CharZero (MvPolynomial (Fin n) ℤ ⧸ m) := by
  sorry
```

Exercise (51). *Let K/F be a simple algebraic extension. Let $K = F(\theta)$. Let L be an intermediate field of K/F. Show that the minimal polynomial of θ over L: $g(x) = x^r + \alpha_1 x^{r-1} + \cdots + \alpha_r$, satisfies that $F(\alpha_1, \ldots, \alpha_r) = L$.*

```
import Mathlib

open IntermediateField

/--
Let $K/F$ be a simple algebraic extension. Let $K = F(\theta)$. Let $L$ be an intermediate field
of $K/F$. Show that the minimal polynomial of $\theta$ over $L$: $g(x) = x^r+\alpha_1x^{r-1} +
\cdots +\alpha_r$, satisfies that $F(\alpha_1, \dots, \alpha_r) = L$.
-/
theorem adjoin_minpoly_coeffs_toSet_eq {F K : Type} [Field F] [Field K]
    [Algebra F K] {θ : K} (L : IntermediateField F K) (h : F⟮θ⟯ = ⊤) :
    IntermediateField.adjoin F ((↑)'' (minpoly L θ).coeffs.toSet : Set K) = L := by
  sorry
```

Exercise (52). *Let q denote a power of a prime p. Show that the extension $\mathbb{F}_q(t^{1/n})$ over $\mathbb{F}_q(t)$ is Galois if and only if $q \equiv 1 \bmod n$.*

```
import Mathlib

open IntermediateField

/--
Let $q$ denote a power of a prime $p$. Show that the extension $\mathbb{F}_q(t^{1/n})$ over
$\mathbb{F}_q(t)$ is Galois if and only if $q \equiv 1 \bmod n$.
-/
theorem isGalois_galoisField_X_pow_iff_modEq_one (p m n : ℕ) (hn : 1 ≤ n) (hm : 1 ≤ m)
    [Fact p.Prime] :  IsGalois (GaloisField p m)⟮(RatFunc.X ^ n : RatFunc (GaloisField p m))⟯
    (RatFunc (GaloisField p m)) ↔ p ^ m ≡ 1 [MOD n] := by
  sorry
```

Exercise (53). *Prove that every field automorphism of $\mathbb{R}$ that fixes $\mathbb{Q}$ is trivial.*

```
import Mathlib

/--
Prove that every field automorphism of $\mathbb{R}$ that fixes $\mathbb{Q}$ is trivial.
-/
theorem real_algEquiv_eq_one (f : ℝ ≃ₐ[ℚ] ℝ) : f = 1 := by
  sorry
```

Exercise (54). *Let $\mathbb{F}_4$ be the field with 4 elements, t a transcendental over $\mathbb{F}_4$, and $F = \mathbb{F}_4(t^4 + t)$ and $K = \mathbb{F}_4(t)$. Show that K is Galois over F.*

```
import Mathlib

open IntermediateField RatFunc

/--
Let $\mathbb{F}_4$ be the field with $4$ elements, $t$ a transcendental over $\mathbb{F}_4$,
and $F =\mathbb{F}_4(t^4 + t)$ and $K =\mathbb{F}_4(t)$. Show that $K$ is Galois over $F$.
-/
theorem isGalois_galoisField_adjoin_X_pow_four_add_X :
    IsGalois (GaloisField 2 2)⟮(X ^ 4 + X : RatFunc (GaloisField 2 2))⟯
    (RatFunc (GaloisField 2 2)) := by
  sorry
```

Exercise (55). *Let K be a field with $\mathrm{char}(K) \neq 2$. Consider a Galois extension L/K. Show that $\mathrm{Gal}(L/K) \cong (\mathbb{Z}/2\mathbb{Z})^2$ if and only if the extensions L/K has the form $L = K(\sqrt{a}, \sqrt{b})$ for $a, b \in K^\times$ such that a, b, and a/b are nonsquares in $K^\times$.*

```
import Mathlib

open IntermediateField

/--
Let $K$ be a field with $\operatorname{char}(K) \neq 2$. Consider a Galois extension $L/K$.
Show that $\operatorname{Gal}(L/K) \cong (\mathbb{Z}/2\mathbb{Z})^2$ if and only if
the extensions $L/K$ has the form $L = K(\sqrt{a}, \sqrt{b})$ for $a, b \in K^\times$ such that
$a$, $b$, and $a/b$ are nonsquares in $K^\times$.
-/
theorem exists_pow_two_eq_algebraMap_and_adjoin_eq_top {K L : Type} [Field K] [Field L]
    [Algebra K L] [IsGalois K L] (hK : ¬ CharP K 2) : IsKleinFour (L ≃ₐ[K] L) ↔
    ∃ a b : Kˣ, ∃ x y : L, x ^ 2 = algebraMap K L a.1 ∧ y ^ 2 = algebraMap K L b.1 ∧
    K⟮x, y⟯ = ⊤ ∧ ¬IsSquare a ∧ ¬IsSquare b ∧ ¬IsSquare (a / b) := by
  sorry
```

Exercise (56). *Prove that for n odd, $n > 1$, $\Phi_{2n}(x) = \Phi_n(-x)$, where Φ_n is the nth cyclotomic polynomial over $\mathbb{Q}$.*

```
import Mathlib

open Polynomial

/--
Prove that for $n$ odd, $n>1$, $\Phi_{2n}(x) = \Phi_n(-x)$, where $\Phi_n$ is the $n$th
cyclotomic polynomial over $\mathbb{Q}$.
-/
theorem cyclotomic_two_mul_eq_cyclotomic_comp_neg {n : ℕ} (hn : Odd n) (hn' : 1 < n) :
    Polynomial.cyclotomic (2 * n) ℚ = (Polynomial.cyclotomic n ℚ).comp (-X) := by
  sorry
```

Exercise (57). *If F is a field that is not perfect, show that F has a nontrivial purely inseparable extension.*

```
import Mathlib

/--
If $F$ is a field that is not perfect, show that $F$ has a nontrivial purely inseparable
extension.
-/
theorem exists_isPurelyInseparable_and_bot_lt {F : Type} [Field F] (h : ¬ PerfectField F) :
    ∃ E : IntermediateField F (AlgebraicClosure F), IsPurelyInseparable F E ∧ ⊥ < E := by
  sorry
```

Exercise (58). *Show that there is at most one extension $F(\alpha)$ of a field F such that $\alpha^4 \in F$, $\alpha^2 \notin F$, and $F(\alpha) = F(\alpha^2)$.*

```
import Mathlib

open IntermediateField
/--
Show that there is at most one extension \( F(\alpha) \) of a field \( F \) such that \( \alpha^4
\in F \), \( \alpha^2 \notin F \), and \( F(\alpha) = F(\alpha^2) \).
-/
theorem exists_eq_adjoin_and_pow_four_mem_bot_and_not_pow_two_mem_bot (F : Type) [Field F] :
    Subsingleton {M : IntermediateField F (AlgebraicClosure F) //
    ∃ α : AlgebraicClosure F, M = F⟮α⟯ ∧ α ^ 4 ∈ (⊥ : IntermediateField F (AlgebraicClosure F)) ∧
    ¬ α ^ 2 ∈ (⊥ : IntermediateField F (AlgebraicClosure F)) ∧ F⟮α⟯ = F⟮α ^ 2⟯} := by
  sorry
```

Exercise (59). *Show that $x^7 - 11$ has no root in the splitting field of $x^7 - 12$ over $\mathbb{Q}$.*

```
import Mathlib

/--
Show that \( x^7 - 11 \) has no root in the splitting field of \(x^7 - 12\) over \(\mathbb{Q}\).
```

```
-/
theorem rootSet_isEmpty_in_splittingField :
    (.X ^ 7 - 11 : Polynomial ℚ).rootSet ((.X ^ 7 - 12 : Polynomial ℚ).SplittingField) = ⊥ := by
  sorry
```

Exercise (60). *For a positive integer a, consider the polynomial*

$$f_a = x^6 + 3ax^4 + 3x^3 + 3ax^2 + 1.$$

Let F be the splitting field of f_a. Show that its Galois group is solvable.

```
import Mathlib

open Polynomial
/--
For a positive integer $a$, consider the polynomial $$ f_a = x^6 + 3ax^4 + 3x^3 + 3ax^2 + 1.
 $$ Let $F$ be the splitting field of $f_a$. Show that its Galois group is solvable.
-/
theorem isSolvable_X_pow_six_add_gal {a : ℤ} (ha : a > 0) : IsSolvable
    (X ^ 6 + C (3 * a : ℚ) * X ^ 4 + C 3 * X ^ 3 + C (3 * a : ℚ) * X ^ 2 + C 1 : ℚ[X]).Gal:= by
  sorry
```

Exercise (61). *Prove that the polynomial $x^4 + 1$ is not irreducible over any field of positive characteristic.*

```
import Mathlib

open Polynomial
/--
Prove that the polynomial $x^4+1$ is not irreducible over any field of positive characteristic.
-/
theorem not_irreducible_X_pow_four_add_one {F : Type} [Field F] {p : ℕ} [Fact p.Prime]
    [CharP F p] : ¬ Irreducible (X ^ 4 + C 1 : F[X]) := by
  sorry
```

Exercise (62). *Let F be a field and let $f(x) \in F[x]$ be an irreducible polynomial with splitting field E over F. Choose $\alpha \in E$ with $f(\alpha) = 0$. Furthermore, for some fixed integer $n \geq 1$, let $g(x)$ be an irreducible polynomial in $F[x]$ with $g(\alpha^n) = 0$. If $\deg(f)/\deg(g) = n$ and if the characteristic of F does not divide n, prove that E contains a primitive nth root of unity.*

```
import Mathlib

/--
Let $F$ be a field and let $f(x) \in F[x]$ be an irreducible polynomial with splitting field $E$ over
    $F$.
Choose $\alpha \in E$ with $f(\alpha) =0$. Furthermore, for some fixed integer $n \geq 1$,
```

```
let $g(x)$ be an irreducible polynomial in $F[x]$ with $g(\alpha^n)=0$. If $\deg(f) / \deg(g) = n$
and if the characteristic of $F$ does not divide $n$, prove that $E$ contains a primitive $n$th root
    of unity.
-/
theorem primitiveRoots_not_empty (E F : Type) [Field E] [Field F] [Algebra F E]
    (f : Polynomial F) (h_f_irr : Irreducible f) (h_splitting_field : f.IsSplittingField F E)
    (a : E) (ha : f.aeval a = 0) (n : ℕ) (hn : n ≥ 1)
    (g : Polynomial F) (h_g_irr : Irreducible g) (h_ga : g.aeval (a ^ n) = 0)
    (h_deg : f.degree = g.degree * n) (h_char : (n : F) ≠ 0) : primitiveRoots n E ≠ ∅ := by
  sorry
```

Exercise (63). *Let $f \in \mathbb{Q}[X]$ and $\xi \in \mathbb{C}$ be a root of unity. Show that $f(\xi) \neq 2^{\frac{1}{4}}$.*

```
import Mathlib

open Polynomial
/--
Let \( f \in \mathbb{Q}[X] \) and \( \xi \in \mathbb{C} \) be a root of unity. Show that \( f(\xi)
    \neq 2^{\frac{1}{4}} \).
-/
theorem eval₂_algebraMap_ne_two_pow_one_dvd_four {n : ℕ} (hn : 1 ≤ n) (f : ℚ[X]) (ξ : ℂˣ)
    (h : ξ ∈ rootsOfUnity n ℂ) : f.eval₂ (algebraMap ℚ ℂ) ξ ≠ (2 : ℂ) ^ ((1 : ℂ) / 4) := by
  sorry
```

Exercise (64). *Let K be a finite extension of a field F, and let $f(x) \in K[x]$. Prove that there exists a nonzero polynomial $g(x) \in K[x]$ such that $f(x)g(x) \in F[x]$.*

```
import Mathlib

open Polynomial
/--
Let $K$ be a finite extension of a field $F$, and let $f(x) \in K[x]$. Prove that there exists a
nonzero polynomial $g(x) \in K[x]$ such that $f(x)g(x) \in F[x]$.
-/
theorem exists_mul_eq_map_algebraMap {F K : Type} [Field F] [Field K] [Algebra F K]
    [FiniteDimensional F K]
    (f : K[X]) : ∃ g ≠ 0, ∃ h : F[X], f * g = h.map (algebraMap F K) := by
  sorry
```

Exercise (65). *Prove that for any $a, b \in \mathbb{F}_{p^n}$ that if $x^3 + ax + b$ is irreducible then $-4a^3 - 27b^2$ is a square in $\mathbb{F}_{p^n}$.*

```
import Mathlib

open Polynomial
/--
Prove that for any $a,b \in \mathbb{F}_{p^n}$ that if $x^3+ax+b$ is irreducible then $-4a^3-27b^2$ is
    a
```

```
 square in $\mathbb{F}_{p^n}$.
-/
theorem isSquare_discriminant_of_irreducible {p n : ℕ} [Fact p.Prime] (a b : GaloisField p n)
    (h_irr : Irreducible (X ^ 3 + C a * X + C b)) :
    IsSquare (- 4 * a ^ 3 - 27 * b ^ 2) := by
  sorry
```

Exercise (66). *Prove that, if $n \geq 3$, then $x^{2^n} + x + 1$ is* reducible *in $\mathbb{F}_2$.*

```
import Mathlib

open Polynomial
/--
Prove that, if $n \geq 3$, then $x^{2^n}+x+1$ is \emph{reducible} in $\mathbb{F}_2$.
-/
theorem not_irreducible_X_pow_two_pow_add_X_add_C_one {n : ℕ} (hn : n ≥ 3) :
    ¬ Irreducible (X ^ 2 ^ n + X + C 1 : (ZMod 2)[X]) := by
  sorry
```

Exercise (67). *Prove that the prime ideals of $\mathbb{F}_7[\alpha] \otimes_{\mathbb{F}_7} \mathbb{F}_7[\alpha]$ are principal, where $\alpha^3 = 2$.*

```
import Mathlib

open Polynomial

/--
Prove that the prime ideals of $\mathbb{F}_7[\alpha] \otimes_{\mathbb{F}_7} \mathbb{F}_7[\alpha]$ are
    principal, where $\alpha^3 = 2$.
-/
theorem isPrincipal_of_ideal_tensor_zMod_seven
    (p : Ideal (TensorProduct (ZMod 7) (AdjoinRoot (X ^ 3 - 2 : Polynomial (ZMod 7)))
    (AdjoinRoot (X ^ 3 - 2 : Polynomial (ZMod 7))))) [p.IsPrime] : p.IsPrincipal := sorry
```

Exercise (68). *Let A be a Noetherian ring and let $x \in A$ be an element which is neither a unit nor a zero-divisor. Prove that the ideals xA and x^nA for $n = 1, 2, \ldots$ have the same prime divisors:*

$$\mathrm{Ass}_A(A/xA) = \mathrm{Ass}_A(A/x^nA).$$

```
import Mathlib

open Ideal
/--
Let $A$ be a Noetherian ring and let $x \in A$ be an element which is
neither a unit nor a zero-divisor. Prove that the ideals $xA$ and $x^nA$
for $n = 1, 2, \dots$ have the same prime divisors:
\[
\operatorname{Ass}_A(A/xA) = \operatorname{Ass}_A(A/x^nA).
\]
```

```
-/
theorem associatedPrimes_quot_span_eq_quot_span_pow {A : Type} [CommRing A] [IsNoetherianRing A]
    (x : A) (hx : x ∈ nonZeroDivisors A) (hx' : ¬ IsUnit x) (n : ℕ) (h : n ≥ 1) :
    associatedPrimes A (A / span {x}) = associatedPrimes A (A / span {x ^ n}) := by
  sorry
```

Exercise (69). *Prove that if K is a field of finite degree over $\mathbb{Q}$ and $x_1, \dots, x_n$ are finitely many elements of K, then the subring $\mathbb{Z}[x_1, \dots, x_n]$ they generate over $\mathbb{Z}$ is not equal to K.*

```
import Mathlib

/--
Prove that if $K$ is a field of finite degree over $\mathbb{Q}$ and $x_1, \dots, x_n$ are
finitely many elements of $K$, then the subring $\mathbb{Z}[x_1, \dots, x_n]$ they generate over
$\mathbb{Z}$ is not equal to $K$.
-/
theorem int_adjoin_range_ne_top {K : Type} [Field K] [NumberField K] {n : ℕ} (x : Fin n → K) :
    Algebra.adjoin ℤ (Set.range x) ≠ ⊤ := by
  sorry
```

Exercise (70). *Show that $\mathbb{Z}[x]/(x^2+4)$ is not integrally closed.*

```
import Mathlib

open Polynomial

/--
Show that $\mathbb{Z}[X]/(x^2+4)$ is not integrally closed.
-/
theorem not_isIntegrallyClosed_adjoinRoot_pow_two_add_four :
    ¬ IsIntegrallyClosed (AdjoinRoot (X ^ 2 + C 4 : ℤ[X])) := by
  sorry
```

Exercise (71). *Prove that $k[x, y]$ is not a Dedekind ring.*

```
import Mathlib

open Polynomial

/--
Prove that $k[x,y]$ is not a Dedekind ring.
-/
theorem not_isDedekindRing_mvPolynomial_fin_two {k : Type} [Field k] :
    ¬ IsDedekindRing (MvPolynomial (Fin 2) k) := by
  sorry
```

Exercise (72). *Let A be a ring such that for each maximal ideal $\mathfrak{m}$ of A, the local ring $A_{\mathfrak{m}}$ is Noetherian; and for each $x \neq 0$ in A, the set of maximal ideals of A which contain x is finite. Show that A is Noetherian.*

```
import Mathlib

/--
Let $A$ be a ring such that for each maximal ideal $\mathfrak{m}$ of $A$, the local ring
$A_{\mathfrak{m}}$ is Noetherian; and for each $x \neq 0$ in $A$, the set of maximal ideals of $A$
which contain $x$ is finite. Show that $A$ is Noetherian.
-/
theorem isNoetherianRing_of_finite_isMaximal_and_mem {A : Type} [CommRing A]
    (h_local : ∀ (m : Ideal A), [m.IsMaximal] → IsNoetherianRing (Localization.AtPrime m))
    (h_finite : ∀ x : A, x ≠ 0 → Set.Finite {m : Ideal A | m.IsMaximal ∧ x ∈ m}) :
    IsNoetherianRing A := by
  sorry
```

Exercise (73). *If R is a valuation ring, then an R-module A is flat if it is torsion-free.*

```
import Mathlib

/--
If $R$ is a valuation ring, then an $R$-module $A$ is flat if it is torsion-free.
-/
theorem flat_of_noZeroSMulDivisor {R A : Type} [CommRing R] [IsDomain R] [ValuationRing R]
    [AddCommGroup A]
    [Module R A] [NoZeroSMulDivisors R A] : Module.Flat R A := by
  sorry
```

Exercise (74). *If R is a valuation ring, then an R-module A is torision-free if it is flat.*

```
import Mathlib

/--
If $R$ is a valuation ring, then an $R$-module $A$ is torsion-free if it is flat.
-/
theorem noZeroSMulDivisors_of_flat {R A : Type} [CommRing R] [IsDomain R] [ValuationRing R]
    [AddCommGroup A]
    [Module R A] [Module.Flat R A] : NoZeroSMulDivisors R A := by
  sorry
```

Exercise (75). *Suppose A and B are commutative rings containing a field k, with B finitely generated over k as a ring. If $\varphi : A \to B$ is a ring homomorphism with $\varphi|_k = Id$ and if $Q \subset B$ is a maximal ideal, prove that $\varphi^{-1}(Q) \subset A$ is a maximal ideal.*

```
import Mathlib

/--
Suppose \( A \) and \( B \) are commutative rings containing a field \( k \), with \( B \)
finitely generated over \( k \) as a ring. If \( \varphi : A \to B \) is a ring homomorphism with
\( \varphi|_k = \text{Id} \) and if \( Q \subset B \) is a maximal ideal, prove that \( \varphi^{-1}
(Q) \subset A \) is a maximal ideal.
-/
theorem comap_isMaximal_of_finiteType {A B k : Type} [CommRing A] [CommRing B] [Field k] [Algebra k
    A] [Algebra k B]
    [Algebra.FiniteType k B] (φ : A →ₐ[k] B) (Q : Ideal B) [hQ : Q.IsMaximal] :
    (Ideal.comap φ Q).IsMaximal := by
  sorry
```

Exercise (76). *Show that a finite torsion-free module over a Dedekind domain is projective.*

```
import Mathlib

/--
Show that a finite torsion-free module over a Dedekind domain is projective.
-/
theorem projective_of_noZeroSMulDivisor {R M : Type} [CommRing R] [AddCommGroup M] [Module R M]
    [IsDedekindDomain R] [Module.Finite R M] [NoZeroSMulDivisors R M] : Module.Projective R M := by
  sorry
```

Exercise (77). *Let k be a field, A a local k-algebra with maximal ideal $\mathfrak{m}$. Assume that A is a localization of a k-algebra R and that $A/\mathfrak{m} = k$. Prove that there exists maximal ideal $\mathfrak{n}$ of R with $R_{\mathfrak{n}} = A$.*

```
import Mathlib

/--
Let \( k \) be a field, \( A \) a local \( k \)-algebra with maximal ideal \( \mathfrak{m} \).
Assume that \( A \) is a localization of a \( k \)-algebra \( R \) and that \( A / \mathfrak{m} = k
\). Prove that there exists maximal ideal \( \mathfrak{n} \) of \( R \) with \( R_{\mathfrak{n}} = A
\).
-/
theorem exists_isMaximal_atPrime_of_bijective {k R A : Type} [Field k] [CommRing R] [CommRing A]
    [Algebra k R]
    [Algebra R A] [Algebra k A] [IsScalarTower k R A] [IsLocalRing A]
    (h : Function.Bijective <| (IsLocalRing.residue A).comp (algebraMap k A))
    (S : Submonoid R) [IsLocalization S A] :
    ∃ n : Ideal R, ∃ hn : n.IsMaximal, IsLocalization.AtPrime A n := by
  sorry
```

Exercise (78). *Let R'/R be an integral extension of rings. Show that $rad(R) = rad(R') \cap R$, where $rad(R)$ denotes the nilpotent radical of R.*

```
import Mathlib

/--
Let \( R' / R \) be an integral extension of rings. Show that \( \text{rad}(R) = \text{rad}(R') \cap
    R \),
where $\text{rad}(R)$ denotes the nilpotent radical of $R$.
-/
theorem nilpotent_eq_contraction_nilpotent_of_integral (R R' : Type) [CommRing R] [CommRing R']
    [Algebra R R'] (int : Algebra.IsIntegral R R') :
    nilradical R = Ideal.comap (algebraMap R R') (nilradical R') := by
  sorry
```

Exercise (79). *Let R be a commutative ring. If all submodules of finitely generated free modules over R are free over R, then R is a PID.*

```
import Mathlib

/--
Let \( R \) be a commutative ring. If all submodules of finitely generated free modules over
\( R \) are free over \( R \), then \( R \) is a PID.
-/
theorem isPrincipalIdealRing_of_forall_free {R : Type} [CommRing R]
    (h : ∀ (M : Type) [AddCommGroup M] [Module R M] [Module.Finite R M] [Module.Free R M],
      ∀ (N : Submodule R M), Module.Free R N) :
    IsPrincipalIdealRing R := by
  sorry
```

Exercise (80). *Let $R \subset R'$ be an extension of integral domains, and let $\overline{R}$ be the integral closure of R in R'. Show that for any two monic polynomials $f, g \in R'[t]$ with $f \cdot g \in R[t]$, we have $f, g \in \overline{R}[t]$.*

```
import Mathlib

open Polynomial

/--
Let \( R \subset R' \) be an extension of integral domains, and let \( \overline{R} \) be the
    integral closure of \( R \) in \( R' \).
Show that for any two monic polynomials \( f, g \in R'[t] \) with \( f \cdot g \in R[t] \),
we have \( f, g \in \overline{R}[t] \).
-/
theorem mem_polynomial_integral_closure_of_prod_mem_polynomial (R S : Type) [CommRing R]
    [IsDomain R] [CommRing S] [IsDomain S] [Algebra R S] [NoZeroSMulDivisors R S] (f g : S[X])
    (mem : f * g ∈ lifts (algebraMap R S)) (hf : f.Monic) (hg : g.Monic) :
    f ∈ lifts (integralClosure R S).subtype ∧ g ∈ lifts (integralClosure R S).subtype := by
  sorry
```

Exercise (81). *Let $R = \mathbb{C}[x_1, \ldots, x_n]/(x_1^2 + x_2^2 + \cdots + x_n^2)$. Then R is not a unique factorization domain for $n = 3$ or 4.*

```
import Mathlib

open MvPolynomial
/--
Let \( R = \mathbb{C}[x_1, \dots, x_n]/(x_1^2 + x_2^2 + \dots + x_n^2) \).
-/
abbrev R (n : ℕ) : Type :=
  MvPolynomial (Fin n) ℂ / Ideal.span {(∑ i : Fin n, X i ^ 2 : MvPolynomial (Fin n) ℂ)}

/--
Let \( R = \mathbb{C}[x_1, \dots, x_n]/(x_1^2 + x_2^2 + \dots + x_n^2) \).
Then \( R \) is not a unique factorization domain for \( n = 3 \) or \( 4 \).
-/
theorem not_UFD_of_3_or_4 (n : ℕ) (h : n = 3 ∨ n = 4) [IsDomain (R n)] :
    ¬ UniqueFactorizationMonoid (R n) := by
  sorry
```

Exercise (82). *Let D be a unique factorization domain. Prove that if every nonzero prime ideal of D is maximal, then D is a principal ideal domain.*

```
import Mathlib

/--
Let \( D \) be a unique factorization domain.
Prove that if every nonzero prime ideal of \( D \) is maximal, then \( D \) is a principal ideal
    domain.
-/
theorem isPrincipalIdealRing_of_isPrime_ne_bot_isMaximal {D : Type} [CommRing D] [IsDomain D]
    [UniqueFactorizationMonoid D] (h : ∀ P : Ideal D, [P.IsPrime] → P ≠ ⊥ → P.IsMaximal) :
    IsPrincipalIdealRing D := by
  sorry
```

Exercise (83). *Let M be a finitely-generated module over a Dedekind domain. Prove that M is flat if and only if M is torsion-free.*

```
import Mathlib


/--
Let $M$ be a finitely-generated module over a Dedekind domain. Prove that $M$ is flat if and
only if $M$ is torsion-free.
-/
theorem flat_iff_noZeroSMulDivisor {R M : Type} [CommRing R] [AddCommGroup M]
    [Module R M] [IsDedekindDomain R] [Module.Finite R M] :
    Module.Flat R M ↔ NoZeroSMulDivisors R M := by
  sorry
```

Exercise (84). *Let A be a Dedekind domain and $\mathfrak{a} \neq 0$ an ideal in A. Show that every ideal in $A/\mathfrak{a}$ is principal.*

```
import Mathlib


/--
Let $A$ be a Dedekind domain and $\mathfrak{a} \neq 0$ an ideal in $A$. Show that every ideal in
 $A/\mathfrak{a}$ is principal.
-/
theorem isPrincipalIdealRing_quot_of_isDedekind {A : Type} [CommRing A]
    [IsDedekindDomain A] (a : Ideal A) (ha : a ≠ ⊥) :
    IsPrincipalIdealRing (A ⧸ a) := by
  sorry
```

Exercise (85). *Let A be a Noetherian ring and let $\mathfrak{a}, \mathfrak{b}$ be ideals in A. If M is any A-module, let $M_{\mathfrak{a}}$, $M_{\mathfrak{b}}$ denote its $\mathfrak{a}$-adic and $\mathfrak{b}$-adic completions respectively. If M is finitely generated, prove that $(M_{\mathfrak{a}})_{\mathfrak{b}} \cong M_{\mathfrak{a}+\mathfrak{b}}$.*

```
import Mathlib


open Submodule
open Finset
open Submodule

/--
Let $A$ be a Noetherian ring and let $\mathfrak{a}, \mathfrak{b}$ be ideals in $A$. If $M$ is
any $A$-module, let $M_{\mathfrak{a}}$, $M_{\mathfrak{b}}$ denote its $\mathfrak{a}$-adic and
$\mathfrak{b}$-adic completions respectively. If $M$ is finitely generated, prove that
$(M_{\mathfrak{a}})_{\mathfrak{b}} \cong M_{\mathfrak{a} + \mathfrak{b}}$.
-/
theorem nonempty_adicCompletion_adicCompletion_linearEquiv {A M : Type} [CommRing A]
    [IsNoetherianRing A] (𝔞 : Ideal A) [AddCommGroup M]
    [Module A M] [Module.Finite A M] (ℬ : Ideal A) :
    Nonempty (AdicCompletion ℬ (AdicCompletion 𝔞 M) ≃ₗ[A] AdicCompletion (𝔞 + ℬ) M) := by
  sorry
```

Exercise (86). *Let M be an R-module. The following are equivalent:*

1. *M is finitely generated.*
2. *For every family $(Q_\alpha)_{\alpha \in A}$ of R-modules, the canonical map $M \otimes_R (\prod_\alpha Q_\alpha) \to \prod_\alpha (M \otimes_R Q_\alpha)$ is surjective.*

```
import Mathlib

/--
Let $M$ be an $R$-module. The following are equivalent:
```

```
\begin{enumerate}
    \item $M$ is finitely generated.
    \item For every family $(Q_{\alpha})_{\alpha \in A}$ of $R$-modules, the canonical map
    $M \otimes_{R} \left( \prod_{\alpha} Q_{\alpha} \right) \to \prod_{\alpha} (M \otimes_{R}
    Q_{\alpha})$ is surjective.
\end{enumerate}
-/
theorem finite_iff_surjective_linearMap {R : Type} [CommRing R]
    (M : Type) [AddCommGroup M] [Module R M] :
    Module.Finite R M ↔ ∀ {α : Type} (Q : α → Type),
    ∀ [(a : α) → AddCommGroup (Q a)] [(a : α) → Module R (Q a)],
        Function.Surjective (LinearMap.pi (
            fun i => LinearMap.lTensor M (
                LinearMap.proj i (φ := Q) (R := R)))) := by
  sorry
```

Exercise (87). *If R is a valuation ring of Krull dimension 1 and K its field of fractions, then there does not exist any rings intermediate between R and K.*

```
import Mathlib

/--
If $R$ is a valuation ring of Krull dimension 1 and $K$ its field of fractions, then there do
not exist any rings intermediate between $R$ and $K$.
-/
theorem eq_bot_or_eq_top_of_ringKrullDim_eq_one {R K : Type} [CommRing R] [IsDomain R]
    [Field K] [Algebra R K] [IsFractionRing R K] [ValuationRing R] (hD : ringKrullDim R = 1)
    (S : Subalgebra R K) : S = ⊥ ∨ S = ⊤ := by
  sorry
```

Exercise (88). *Let $(R, \mathfrak{m})$ be a Noetherian local ring. Let $x, y \in \mathfrak{m}$ be a regular sequence of length 2. For any $n \geq 2$ show that there do not exist $a, b \in R$ with*

$$x^{n-1}y^{n-1} = ax^n + by^n$$

```
import Mathlib

open RingTheory
/--
Let \((R, \mathfrak{m})\) be a Noetherian local ring. Let \(x, y \in \mathfrak{m}\) be a regular
sequence of length \(2\). For any \(n \geq 2\) show that there do not exist \(a, b \in R\) with
\[
x^{n-1}y^{n-1} = a x^{n} + b y^{n}
\]
-/
theorem not_exists_pow_sub_one_mul_pow_sub_one_eq {R : Type} [CommRing R] [IsNoetherianRing R]
    [IsLocalRing R] {n : ℕ} {x y : R} (hn : n ≥ 2) (hx : x ∈ IsLocalRing.maximalIdeal R)
    (hy : y ∈ IsLocalRing.maximalIdeal R) (h : Sequence.IsRegular R [x, y]) :
```

```
    ¬ ∃ a b : R, x ^ (n - 1) * y ^ (n - 1) = a * x ^ n + b * y ^ n := by
  sorry
```

Exercise (89). *Let K be a field and L an extension field of K. If P is a prime ideal of $L[X_1, \ldots, X_n]$ and $\mathfrak{p} = P \cap K[X_1, \ldots, X_n]$, then* $\mathrm{ht}(P) \geq \mathrm{ht}(\mathfrak{p})$.

```
import Mathlib

open MvPolynomial
/--
Let \( K \) be a field and \( L \) an extension field of \( K \). If \( P \) is a prime ideal of
\( L[X_1, \dots, X_n] \) and \( \mathfrak{p} = P \cap K[X_1, \dots, X_n] \), then \(
\operatorname{ht}(P) \geq \operatorname{ht}(\mathfrak{p}) \).
-/
theorem primeHeight_le_of_comap_eq {K L : Type} (n : ℕ) [Field K] [Field L] [Algebra K L]
    (P : Ideal (MvPolynomial (Fin n) L)) (p : Ideal (MvPolynomial (Fin n) K)) [P.IsPrime]
    [p.IsPrime] (h : P.comap (MvPolynomial.map (algebraMap K L)) = p) :
    p.primeHeight ≤ P.primeHeight := by
  sorry
```

Exercise (90). *Suppose that R is a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field κ. In this case the projective dimension of κ is* $\geq \dim_\kappa \mathfrak{m}/\mathfrak{m}^2$.

```
import Mathlib

/--
Suppose that \( R \) is a Noetherian local ring with maximal ideal \( \mathfrak{m} \) and residue
field \( \kappa \). In this case the projective dimension of \( \kappa \) is
\( \geq \dim_{\kappa} \mathfrak{m} / \mathfrak{m}^{2} \).
-/
theorem not_hasProjectiveDimensionLT_finrank_cotangentSpace {R : Type} [CommRing R] [IsLocalRing R]
    [IsNoetherianRing R] :
      ¬ CategoryTheory.HasProjectiveDimensionLT
        (ModuleCat.of R (IsLocalRing.ResidueField R))
          (Module.finrank (IsLocalRing.ResidueField R) (IsLocalRing.CotangentSpace R)) := by
  sorry
```

Exercise (91). *If a ring R, not a field, is a maximal proper subring of a field K, show R is a valuation ring of Krull dimension 1.*

```
import Mathlib

/--
If a ring \( R \), not a field, is a maximal proper subring of a field \( K \), show \( R \) is
a valuation ring of Krull dimension 1.
-/
theorem exists_toSubring_eq_and_ringKrullDim_eq_one {K : Type} [Field K] (R : Subring K)
```

```
    (h : IsCoatom R) (hR : ¬ IsField R) :
    (∃ V : ValuationSubring K, V.toSubring = R) ∧ ringKrullDim R = 1 := by
  sorry
```

Exercise (92). *Let R be a Dedekind domain. Show the following:*

If $P_1, \dots, P_n \in \operatorname{Spec}(R)$ are pairwise distinct, non-zero prime ideals and $e_1, \dots, e_n$ are non-negative integers, there exists $a \in R \setminus \{0\}$ such that

$$(a) = P_1^{e_1} \cdots P_n^{e_n} \cdot J,$$

with $J \subseteq R$ an ideal in whose factorization none of the P_i appear.

```
import Mathlib

/--
Let \( R \) be a Dedekind domain. Show the following:

If \( P_1, \dots, P_n \in \operatorname{Spec}(R) \) are pairwise distinct, non-zero prime ideals
and \( e_1, \dots, e_n \) are non-negative integers, there exists \( a \in R \setminus \{0\} \)
such that
\[
(a) = P_1^{e_1} \cdots P_n^{e_n} \cdot J,
\]
with \( J \subseteq R \) an ideal in whose factorization none of the \( P_i \) appear.
-/
theorem exists_factor_principal_ideal (R : Type) [CommRing R] [IsDedekindDomain R]
    (n : ℕ) (p : (Fin n) → PrimeSpectrum R) (h_nonzero : ∀ i, (p i).asIdeal ≠ ⊥)
    (h_inj : Function.Injective p) (e : (Fin n) → ℕ) :
    ∃ (a : R) (J : Ideal R), a ≠ 0 ∧ Ideal.span {a} = J * ∏ (i : Fin n), (p i).1 ^ (e i) ∧
    (∀ (i : Fin n), ¬ ∃ (K : Ideal R), J = K * (p i).1) := by
  sorry
```

Exercise (93). *Let $\mathcal{O}$ be an integral domain in which all nonzero ideals admit a unique factorization into prime ideals. Show that $\mathcal{O}$ is a Dedekind domain.*

```
import Mathlib

/--
Let $\mathcal{O}$ be an integral domain in which all nonzero ideals admit a unique factorization
into prime ideals. Show that $\mathcal{O}$ is a Dedekind domain.
-/
theorem isDedekindDomain_of_ideal_UFD (O : Type) [CommRing O] [IsDomain O]
(eif : ∀ (a : Ideal O), a ≠ ⊥ → ∃ (f : Multiset (Ideal O)), (∀ b ∈ f, b.IsPrime) ∧ f.prod = a)
(uif : ∀ (f g : Multiset (Ideal O)), (∀ x ∈ f, x.IsPrime) → (∀ x ∈ g, x.IsPrime) → f.prod ≠ ⊥ →
    f.prod = g.prod → f = g) :
IsDedekindDomain O := by
sorry
```

Exercise (94). *Suppose that a ring S is integral over the image of a ring homomorphism $R \to S$. Show that the Krull dimension of M as an S-module is the same as the Krull dimension of M as an R-module.*

```
import Mathlib

/--
Suppose that a ring $S$ is integral over the image of a ring homomorphism $R \to S$. Show that
the Krull dimension of $M$ as an $S$-module is the same as the Krull dimension of $M$ as an
$R$-module.
-/
theorem ringKrullDim_quot_annihilator_eq {R S M : Type} [CommRing R] [CommRing S] [AddCommGroup M]
    [Algebra R S]
    [Module S M] [Module R M] [IsScalarTower R S M] [Algebra.IsIntegral R S] :
    ringKrullDim (S ⧸ (Module.annihilator S M)) = ringKrullDim (R ⧸ (Module.annihilator R M)) := by
  sorry
```

Exercise (95). *Show that if $x_1, \dots, x_r$ is a regular sequence in R, then so is $x_1^{a_1}, \dots, x_r^{a_r}$ for any positive integers $a_1, \dots, a_r$.*

```
import Mathlib

open RingTheory

/--
Show that if $x_1, \dots, x_r$ is a regular sequence in $R$,
then so is $x_1^{a_1}, \dots, x_r^{a_r}$ for any positive integers $a_1, \dots, a_r$.
-/
theorem isRegular_ofFn_pow (R M : Type) [CommRing R] [AddCommGroup M] [Module R M]
    (rs : List R) (a : Fin rs.length → ℕ+) (h : Sequence.IsRegular M rs) :
    Sequence.IsRegular M (List.ofFn (fun i => rs[i] ^ (a i).1)) := by
  sorry
```

Exercise (96). *Let $I_1, I_2 \subseteq K[x_1, \dots, x_n]$ be two ideals. With y an additional indeterminate, form the ideal*

$$J := \big(y \cdot I_1 \cup (1-y) \cdot I_2\big) K[x_1, \dots, x_n, y] \subseteq K[x_1, \dots, x_n, y].$$

Show that

$$I_1 \cap I_2 = K[x_1, \dots, x_n] \cap J.$$

```
import Mathlib

open Polynomial
/--
Let \( I_1, I_2 \subseteq K[x_1, \dots, x_n] \) be two ideals. With \( y \) an additional
indeterminate, form the ideal
\[
```

```
J := \big( y \cdot I_1 \cup (1 - y) \cdot I_2 \big) K[x_1, \dots, x_n, y] \subseteq K[x_1, \dots,
x_n, y].
\]
Show that
\[
I_1 \cap I_2 = K[x_1, \dots, x_n] \cap J.
\]
-/
theorem inf_eq_span_X_smul_sup_one_sub_X_smul_comap_C {K : Type} [Field K] (n : ℕ)
    (I1 I2 : Ideal (MvPolynomial (Fin n) K)) :
    I1 ⊓ I2 = (Ideal.span {(X : (MvPolynomial (Fin n) K)[X])} • (I1.map C) ⊔
    Ideal.span {(1 - X : (MvPolynomial (Fin n) K)[X])} • (I2.map C) ).comap C := by
  sorry
```

Exercise (97). *Prove that* $\sin 1^\circ$ *is algebraic over* $\mathbb{Q}$.

```
import Mathlib

open Real

/--
Prove that $\sin 1^{\circ}$ is algebraic over $\mathbb{Q}$.
-/
theorem isAlgebraic_sin_pi_div_180 : IsAlgebraic ℚ (sin (π / 180)) := by
  sorry
```

Exercise (98). *Let* A *be a Noetherian ring, and suppose that* P *is a height* $r > 0$ *prime ideal generated by* r *elements, which may* ***not*** *be a regular sequence. Show that* P *can be generated by an* A*-sequence.*

```
import Mathlib

open RingTheory
/--
Let \( A \) be a Noetherian ring, and suppose that \( P \) is a height \( r > 0 \) prime ideal
generated by \( r \) elements, which may **not** be a regular sequence. Show that \( P \) can be
generated by an \( A \)-sequence.
-/
theorem exists_isRegular_and_eq_ofList {A : Type} [CommRing A] [IsNoetherianRing A] (P : Ideal A)
    [P.IsPrime] (r : ℕ)
    (hr : 0 < r) (h : r = P.primeHeight) (a : Fin r → A) (hP : P = Ideal.span (Set.range a)) :
    ∃ b : List A, Sequence.IsRegular A b ∧ P = Ideal.ofList b := by
  sorry
```

Exercise (99). *For a commutative ring* R *and* $n \in \mathbb{N}$ *such that* 2 *is invertible in* R*. If* $A \in SO(2n+1, R)$*, then* $det(I - A) = 0$.

```
import Mathlib

/--
For a commutative ring \(R\) and \(n \in \mathbb{N}\) such that \(2\) is invertible in \(R\).
If \(A \in SO(2n + 1,  R)\), then \(det(I - A) = 0\).
-/
theorem determinant_eq_zero (R : Type) [CommRing R] (n : ℕ) (h2_inv : IsUnit (2 : R))
    (A : Matrix.specialOrthogonalGroup (Fin (2 * n + 1)) R) : (1 - A.1).det = 0 := by
  sorry
```

Exercise (100). *There exists a commutative ring with finite prime spectrum but is not Noetherian.*

```
import Mathlib

/--
There exists a commutative ring with finite prime spectrum but is not Noetherian.
-/
theorem exists_finite_primeSpectrum_not_isNoetherianRing :
    ∃ (R : Type) (_ : CommRing R), Finite (PrimeSpectrum R) ∧ ¬ IsNoetherianRing R := by
  sorry
```